

Hindmarsh-Rose mHAVOK Final Results

80k-sample burst-warning sweep

Question

- Can a multichannel HAVOK embedding produce a forcing-like signal that warns us before HR burst onset?
- Which observable set actually carries the cleanest warning information?

Bottom line

- **Observable choice matters more than channel count.**
- On the tested grid, **z only** is the best warning model.
- The full-state **x+y+z** model is useful, but it is **not the winner**.

POST-TRANSIENT SAMPLES

80,000

BURST ONSETS

39

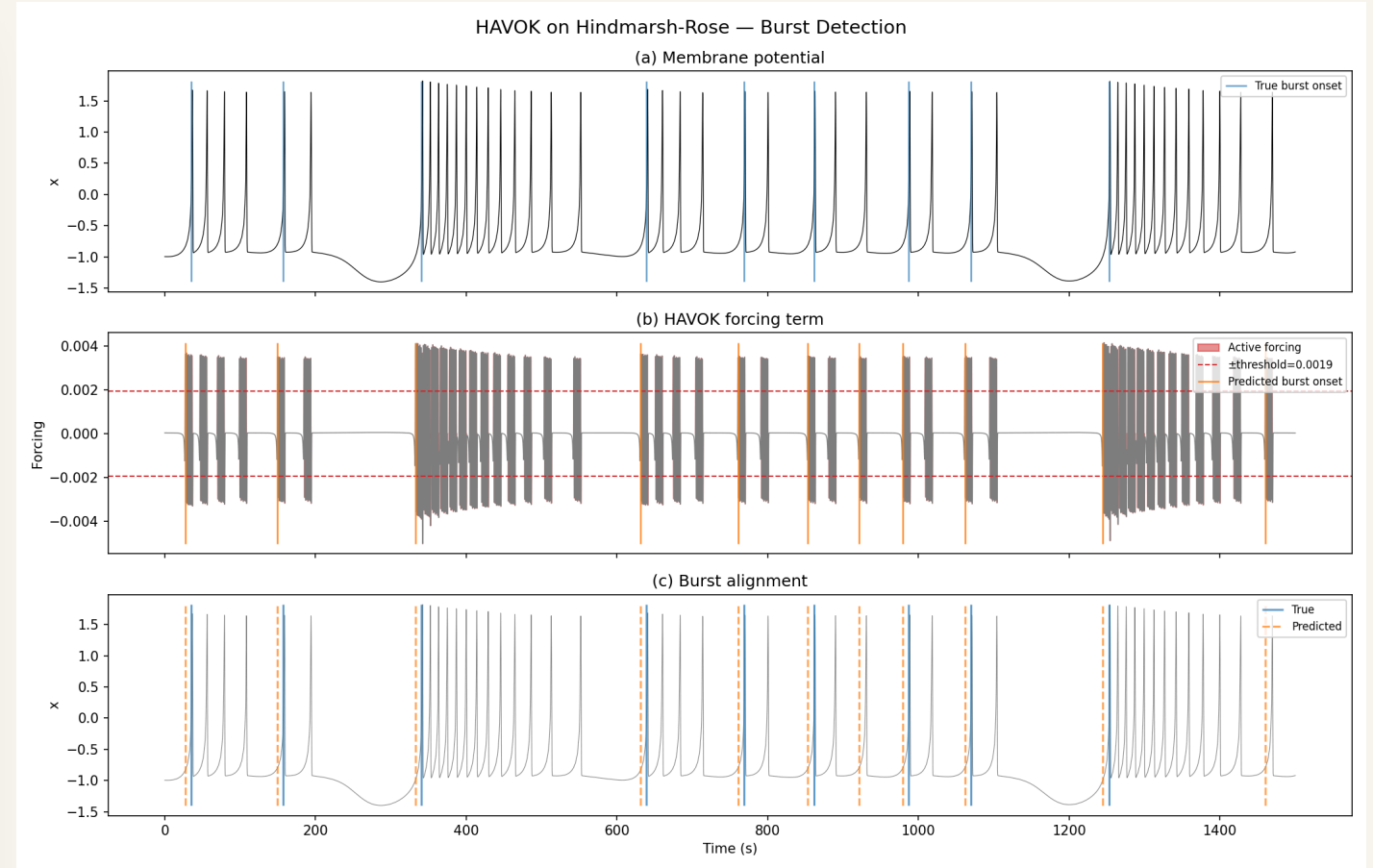
BEST WARNING COMBO

z only

Prediction Target

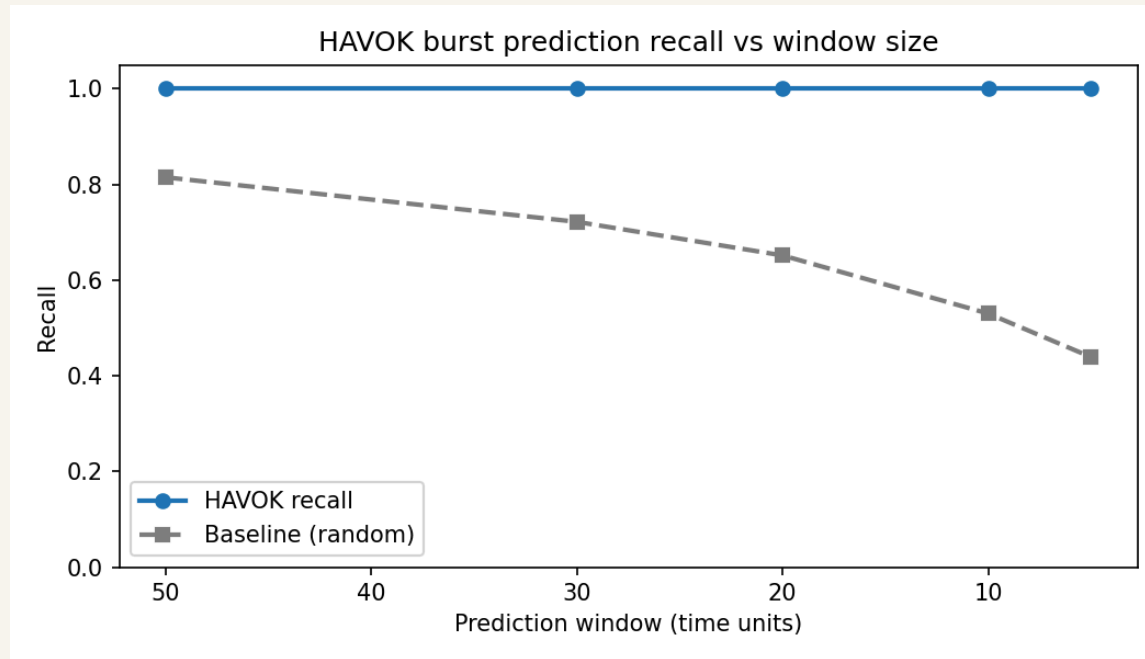
What we want to predict

- The target is **burst onset**, not generic large-amplitude activity.
- A useful warning signal must turn on **before** the burst begins.
- Single-channel HAVOK already shows that forcing activation clusters near burst onset.
- The mHAVOK question is whether **multiple observables** sharpen that warning or

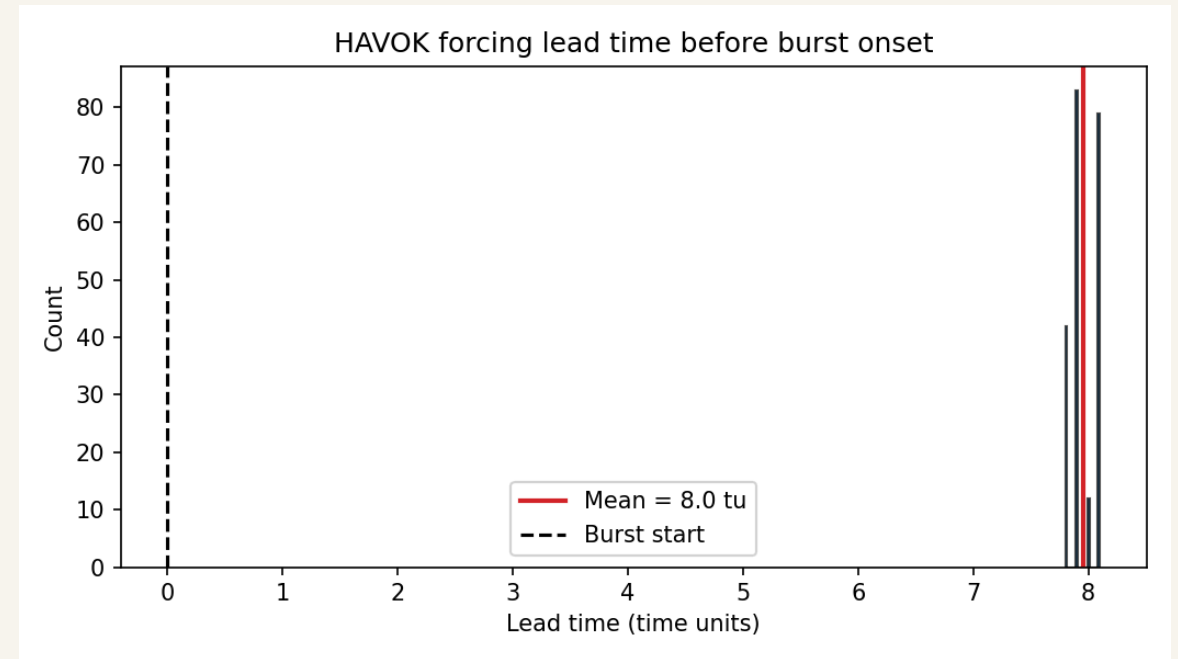


Reference HR burst-detection plot: the presentation target is the onset of each burst, and the forcing-like signal is useful only if it activates beforehand.

Single-Channel HAVOK Baseline



The original HAVOK warning signal already captures every true burst across 50, 30, 20, and 10 time-unit windows.



Those activations are not just present; they arrive before onset, with a mean lead time of about 8.0 time units.

Why this matters for the final results

- This establishes the baseline scientific fact: the forcing idea is already useful as a warning signal in HR.

What mHAVOK Does

1. Observe channels

- Start from the measured HR coordinates: **$x(t)$, $y(t)$, $z(t)$** .
- The goal is to ask which of these channels carries burst-warning information.

2. Build stacked Hankel blocks

- Form one delay-embedding Hankel matrix per channel.
- Stack those blocks vertically so the SVD sees **time history across channels**.

3. Extract latent coordinates

- Use the SVD to find low-dimensional coordinates.
- Treat the last retained coordinate as the **forcing-like mode**.

4. Convert forcing into warnings

- Threshold the forcing signal.
- Merge nearby activations into one event.
- Score whether each predicted event lands inside a pre-burst warning window.

Full Sweep Setup

Sweep design

- Post-transient analysis window: **80,000** samples
- Detected burst onsets: **39**
- Delay grid: **50, 100, 150**
- Rank grid: **5, 7, 9, 11, 13**
- Observable sets: **x, y, z, x+y, x+z, y+z, x+y+z**

What the sweep is testing

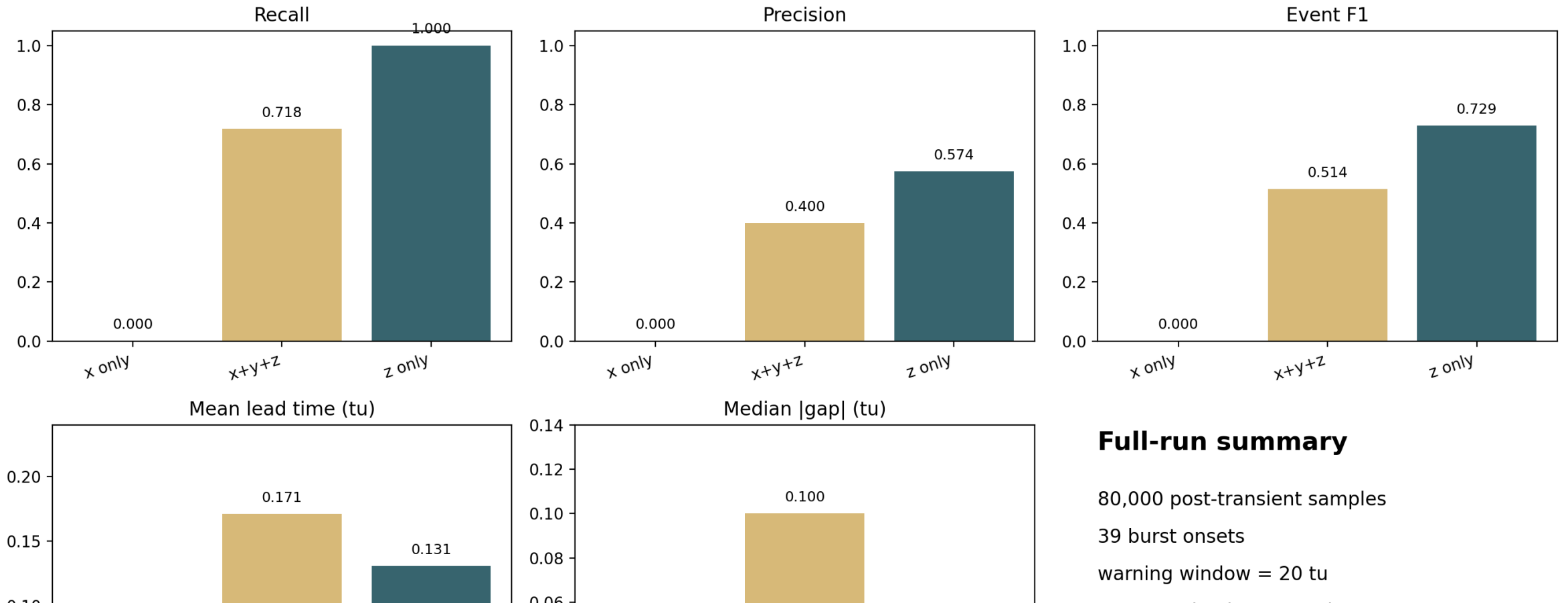
- Does adding channels improve burst-warning quality?
- Does the full-state model **x+y+z** beat simpler observable sets?
- Is the scientifically useful result “more channels help,” or is it “the right channel matters most”?

Evaluation rule

- A model is good when it produces **high recall and precision** for warning events before burst onset.

Main Results: Observable Choice Dominates

Hindmarsh-Rose mHAVOK full-run comparison



What We Can Claim Safely

Defensible claim

- HR mHAVOK is useful as an **observable-screening tool**.
- It shows that burst-warning quality depends strongly on **which channels are included**.
- On this grid, **z** and not **x+y+z** is the best warning input.

Important limit

- Do **not** oversell this as “mHAVOK beats HAVOK everywhere.”
- The strongest direct warning result in the repo is still the original single-channel HAVOK story.
- The multichannel result is valuable because it clarifies **which observables carry warning information**.

BEST Z-ONLY F1

0.729

BEST X+Y+Z F1

0.514

BEST X-ONLY F1

0.000

Presentation Takeaways

How to present this

- Lead with the **prediction problem**: warn before burst onset.
- Spend time explaining what the forcing-like coordinate means.
- Show one onset plot and two summary graphs; keep the dense sweep behind GitHub or a QR code.

Scientific takeaway

- The useful HR mHAVOK result is **not** “more channels are better.”
- The useful result is that **channel choice is a first-order modeling decision**.
- For the tested 80k-sample sweep, the cleanest warning signal comes from **z**.

One-sentence conclusion

- In Hindmarsh-Rose, the forcing-based warning idea remains strong, and the new mHAVOK sweep shows that **the slow adaptation variable is the most informative warning observable on the tested grid**.